1 Theoretical Justification for PCA Combining with Over Selection

This section presents a theoretical result that provides a lower bound on the probability that the true top-k nearest neighbors (in the full d-dimensional space) are captured within the top-wk candidate set identified using only the first d' principal components. This supports the rationale behind the algorithm's Stages 1 and 2.

Remark 1.1 (Notation Mapping)**.** The following theorem uses notation common in statistical analysis. The key mapping to the algorithm description is:

- d: Full dimension (same).
- d': Reduced dimension (referred to as dd in the original theorem source, replaced here by d' for consistency).
- $S_{1,i}$: Squared Euclidean distance between point p_i and query q in the full d-dimensional PCA space. This corresponds to the distance $D_2(\boldsymbol{a}''_j, \boldsymbol{q}''_i)^2$ which determines the true ranking.
- $S_{2,i}$: Squared Euclidean distance between point p_i and query q using only the first d' dimensions of the PCA space. This corresponds to the distance $D_2(\boldsymbol{a}'_j, \boldsymbol{q}'_i)^2$ used for ranking in Stage 1.
- R_i: The residual squared distance, $R_i = S_{1,i} - S_{2,i}$, using dimensions $d'+1$ to d.
- T_1: The set of indices of the true top-k neighbors based on $S_{1,i}$.
- T_2: The set of indices of the top-wk candidates based on $S_{2,i}$ (from Stage 1).
- $\mathbb{P}\{T_1 \subseteq T_2\}$: The probability that the candidate set T_2 contains all the true nearest neighbors T_1. This is closely related to achieving perfect Recall@k within the candidate set, which is a prerequisite for high final recall after Stage 3.

The theorem assumes data follows a multivariate normal distribution, a common simplifying assumption for theoretical analysis.

Theorem 1.2 (Lower Bound for Subset Probability - Multi-Threshold Refinement)**.** *Let data points be drawn from a multivariate normal distribution $N(\mu, \Sigma)$. After centering, the distribution is $X_0 \sim N(0, \Sigma)$. A full-dimensional Principal Component Analysis (PCA) projection yields $X \sim \mathcal{N}(0, \Lambda)$, where $\Lambda = Diag(\lambda_1, \ldots, \lambda_d)$ contains the eigenvalues of Σ in non-increasing order ($\lambda_1 \geq \cdots \geq \lambda_d > 0$).*

Let $p_1, \ldots, p_n, q$ be $n+1$ independent draws from X, where n is large. Define the squared Euclidean distances:

$$S_{2,i} = \sum_{j=1}^{d'} (p_{ij} - q_j)^2, \quad \textit{for } 0 < d' < d \quad (\textit{Reduced dimension distance})$$

$$S_{1,i} = \sum_{j=1}^{d} (p_{ij} - q_j)^2 = S_{2,i} + R_i, \quad \textit{where } R_i = \sum_{j=d'+1}^{d} (p_{ij} - q_j)^2 \quad (\textit{Full distance})$$

Let $S_{2,r_1} \leq S_{2,r_2} \leq \cdots \leq S_{2,r_n}$ be the ordered statistics of $\{S_{2,i}\}_{i=1}^n$. Let $S_{1,t_1} \leq S_{1,t_2} \leq \cdots \leq S_{1,t_n}$ be the ordered statistics of $\{S_{1,i}\}_{i=1}^n$. Define the index sets $T_2 = \{r_1, \ldots, r_{wk}\}$ and $T_1 = \{t_1, \ldots, t_k\}$ for fixed integers k and w, with $wk \leq n$.

Define Approximations and Constants:

1. ***For*** $R_1 = \sum_{j=d'+1}^{d}(p_{1j} - q_j)^2$ ***(Residual Distance Component):***

   - *Moments:* $\mu_R = \mathbb{E}[R_1] = \sum_{j=d'+1}^{d} 2\lambda_j$, $\sigma_R^2 = Var(R_1) = \sum_{j=d'+1}^{d} 8\lambda_j^2$. *Assume* $\sigma_R > 0$.
   - *Satterthwaite-Welch (SW) Approximation:* $R_1 \approx \theta_R \chi^2_{\nu_R}$.
   - *SW parameters:* $\nu_R = 2\mu_R^2/\sigma_R^2$, $\theta_R = \sigma_R^2/(2\mu_R)$.
   - *Approximate CDF:* $F_R(t) := \mathbb{P}(R_1 \le t) \approx F_{\chi^2_{\nu_R}}(t/\theta_R) = P(\nu_R/2, t/(2\theta_R))$, *where* $P(s,x)$ *is the regularized lower incomplete gamma function.*

2. ***For*** $Z(q) = \boldsymbol{Var}(S_{2,i}|q) = \sum_{j=1}^{d'}(2\lambda_j^2 + 4\lambda_j q_j^2)$ ***(Conditional Variance):***

   - *Let* $Z'(q) = Z(q) - \sum_{j=1}^{d'} 2\lambda_j^2 = \sum_{j=1}^{d'} 4\lambda_j^2(q_j^2/\lambda_j)$.
   - *Moments of* $Z'(q)$*:* $\mu_{Z'} = \mathbb{E}_q[Z'(q)] = \sum_{j=1}^{d'} 4\lambda_j^2$, $\sigma_{Z'}^2 = Var_q(Z'(q)) = \sum_{j=1}^{d'} 32\lambda_j^4$.
   - *SW Approximation:* $Z'(q) \approx \theta_{Z'}\chi^2_{\nu_{Z'}}$.
   - *SW parameters:* $\nu_{Z'} = 2\mu_{Z'}^2/\sigma_{Z'}^2$, $\theta_{Z'} = \sigma_{Z'}^2/(2\mu_{Z'})$.
   - *Approximate Survival Function for* $Z(q)$*:* $\bar{F}_Z(y) := \mathbb{P}_q(Z(q) \ge y) = \mathbb{P}_q(Z'(q) \ge y - \sum_{j=1}^{d'} 2\lambda_j^2) \approx Q(\nu_{Z'}/2, \max(0, y - \sum_{j=1}^{d'} 2\lambda_j^2)/(2\theta_{Z'}))$, *where* $Q(s,x)$ *is the regularized upper incomplete gamma function.*

3. ***Order Statistic and Thresholding Constants:***

   - *Quantiles:* $\alpha_k = k/(n+1)$, $\beta_{wk} = wk/(n+1)$.
   - *Normal-distribution based constants: Let* Φ *and* ϕ *be the standard normal CDF and PDF, respectively.*

$$C_{\alpha\beta} = \Phi^{-1}(\beta_{wk}) - \Phi^{-1}(\alpha_k).$$
$$C_\sigma^2 = \frac{\alpha_k(1-\alpha_k)}{\phi(\Phi^{-1}(\alpha_k))^2} + \frac{\beta_{wk}(1-\beta_{wk})}{\phi(\Phi^{-1}(\beta_{wk}))^2} - \frac{2\alpha_k(1-\beta_{wk})}{\phi(\Phi^{-1}(\alpha_k))\phi(\Phi^{-1}(\beta_{wk}))}.$$

4. ***Partition Parameters:***

   - *Choose an integer* $z \ge 2$.
   - *Choose threshold parameters* $y_1, y_2, \dots, y_{z-1} \in \mathbb{R}$ *such that* $-\mu_R/\sigma_R \le y_1 < y_2 < \cdots < y_{z-1}$.
   - *Define thresholds* $d_i = \mu_R + y_i\sigma_R$ *for* $i = 1, \dots, z-1$. *Note* $d_1 \ge 0$.
   - *Define* $d_0 = 0$ *and* $d_z = +\infty$.

5. ***Step 5 Parameters:***

   - *Choose parameters* $\delta_{i,1}, \delta_{i,2} \in \mathbb{R} \cup \{-\infty, +\infty\}$ *such that* $\delta_{i,1} \le \delta_{i,2}$ *for each* $i = 1, \dots, z-1$.
   - *Define* $\eta_i = \Phi(\delta_{i,2}) - \Phi(\delta_{i,1}) \ge 0$. *(Note:* $\Phi(\infty) = 1, \Phi(-\infty) = 0$*).*
   - *Define* $C_{tot}(\delta) = C_{\alpha\beta} + \delta n^{-1/2} C_\sigma$. *Assume parameters are chosen such that* $C_{tot}(\delta_{i,1}) > 0$ *and* $C_{tot}(\delta_{i,2}) > 0$ *whenever* $\delta_{i,1}, \delta_{i,2}$ *are finite and involved in denominators.*

Assumption 1.3 (Regularity Conditions)**.** We assume conditions hold that allow for:

- Large-n asymptotic results for order statistics.
- Validity of the Satterthwaite-Welch approximations for R_1 and $Z(q)$.

- Approximation of the conditional distribution $D_{w,k}|q$ as normal for the purpose of analyzing conditional probabilities, as detailed in Step 5.

Under these conditions, the probability $\mathbb{P}\{T_1 \subseteq T_2\}$ *is bounded below by:*

$$\mathbb{P}\{T_1 \subseteq T_2\} \gtrsim \sum_{i=1}^{z-1} [F_R(d_i)]^k \, \eta_i \left[\bar{F}_Z\left(\frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2}\right) - \bar{F}_Z\left(\frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2}\right)\right] \tag{1}$$

where the term $\bar{F}_Z(d_{i+1}^2/[C_{tot}(\delta_{i,2})]^2)$ *is taken as 0 if* $d_{i+1} = \infty$*. The expression requires the arguments of* $\bar{F}_Z$ *to be non-negative and well-defined.*

Special Case (z=2): *If we choose* $z = 2$*, the sum has only the* $i = 1$ *term. We have* $d_1 = \mu_R + y_1\sigma_R$ *and* $d_2 = \infty$*. The second* $\bar{F}_Z$ *term becomes* $\bar{F}_Z(\infty) = 0$*. Based on the derivation (Step 5, Analyzing* $C_i(q)$*), when* $d_{i+1} = \infty$*, the condition* $C_i(q)$ *simplifies. This corresponds to setting* $\delta_{i,2} = +\infty$ *to obtain a meaningful lower bound on the conditional probability. With* $\delta_{1,2} = +\infty$*, we have* $\eta_1 = \Phi(+\infty) - \Phi(\delta_{1,1}) = 1 - \Phi(\delta_{1,1})$*. Let* $y_1 = y$ *and* $\delta_{1,1} = \delta$*. The bound* (1) *simplifies to:*

$$\mathbb{P}\{T_1 \subseteq T_2\} \gtrsim [F_R(\mu_R + y\sigma_R)]^k [1 - \Phi(\delta)] \bar{F}_Z\left(\frac{(\mu_R + y\sigma_R)^2}{[C_{tot}(\delta)]^2}\right) \tag{2}$$

where $C_{tot}(\delta) = C_{\alpha\beta} + \delta n^{-1/2} C_\sigma$*. The parameters* y, δ *must be chosen such that* $d_1 = \mu_R + y\sigma_R \geq 0$ *and* $C_{tot}(\delta) > 0$*.*

Proof of Theorem 1.2. We proceed in several steps, establishing a sufficient condition for $T_1 \subseteq T_2$, decomposing the probability of this condition, and then bounding the resulting terms using approximations.

Step 1: Problem Setup and Notation

We are given $n+1$ independent points $p_1, \ldots, p_n, q$ drawn from $\mathcal{N}(0, \Lambda)$, where $\Lambda = \mathrm{Diag}(\lambda_1, \ldots, \lambda_d)$ with $\lambda_1 \geq \cdots \geq \lambda_d > 0$. The squared Euclidean distances are defined as: $S_{1,i} = \sum_{j=1}^{d} (p_{ij} - q_j)^2$, $S_{2,i} = \sum_{j=1}^{d'} (p_{ij} - q_j)^2$, and $R_i = S_{1,i} - S_{2,i} = \sum_{j=d'+1}^{d} (p_{ij} - q_j)^2$. The sets $(S_{1,i}, S_{2,i}, R_i)$ for $i = 1, \ldots, n$ are identically distributed given q. The pairs (p_i, q) are i.i.d. across i.

Let $S_{2,r_1} \leq S_{2,r_2} \leq \cdots \leq S_{2,r_n}$ and $S_{1,t_1} \leq S_{1,t_2} \leq \cdots \leq S_{1,t_n}$ be the ordered statistics, defining the index sets $T_2 = \{r_1, \ldots, r_{wk}\}$ and $T_1 = \{t_1, \ldots, t_k\}$. Our goal is to find a lower bound for $\mathbb{P}\{T_1 \subseteq T_2\}$.

Step 2: Establishing the Core Inequality Condition

Let $r_1, \ldots, r_n$ be the indices ordered by increasing $S_{2,i}$ values. Define:

- $R_m = \max\{R_{r_1}, \ldots, R_{r_k}\}$, the maximum residual distance among the k points with the smallest S_2 distances.
- $D_{w,k} = S_{2,r_{wk}} - S_{2,r_k}$, the difference between the wk-th and k-th smallest S_2 distances.

Consider the event $Q := \{D_{w,k} > R_m\}$, which is equivalent to $S_{2,r_{wk}} > S_{2,r_k} + R_m$. We show that if event Q occurs, then $T_1 \subseteq T_2$.

Assume event Q occurs. Let $i \in \{1, \ldots, k\}$ (so $r_i \in \{r_1, \ldots, r_k\}$) and let $j \geq wk + 1$ (so $r_j \notin T_2 = \{r_1, \ldots, r_{wk}\}$). We need to show that $S_{1,r_j} > S_{1,r_i}$.

By the ordering of S_2 values:

- $j \geq wk + 1 \implies S_{2,r_j} \geq S_{2,r_{wk}}$
- $i \leq k \implies S_{2,r_i} \leq S_{2,r_k}$

By the definition of R_m:

- $i \leq k \implies R_{r_i} \leq R_m$

Now, compare S_{1,r_j} and S_{1,r_i}:

$$S_{1,r_j} = S_{2,r_j} + R_{r_j} \geq S_{2,r_{wk}} + R_{r_j}$$

Using the condition Q ($S_{2,r_{wk}} > S_{2,r_k} + R_m$):

$$S_{1,r_j} > (S_{2,r_k} + R_m) + R_{r_j}$$

Now consider S_{1,r_i}:

$$S_{1,r_i} = S_{2,r_i} + R_{r_i} \leq S_{2,r_k} + R_m$$

Comparing the inequalities:

$$S_{1,r_j} > (S_{2,r_k} + R_m) + R_{r_j} \geq S_{1,r_i} + R_{r_j}$$

Since $R_{r_j} = \sum_{l=d'+1}^{d} (p_{r_j,l} - q_l)^2 \geq 0$, we have $S_{1,r_j} > S_{1,r_i}$.

This holds for any $i \leq k$ and any $j \geq wk + 1$. This implies that none of the indices $\{r_{wk+1}, \ldots, r_n\}$ can be among the k smallest S_1 indices (T_1). Therefore, all indices in T_1 must come from the set $\{r_1, \ldots, r_{wk}\} = T_2$. Thus, $T_1 \subseteq T_2$.

It follows that:

$$\mathbb{P}\{T_1 \subseteq T_2\} \geq \mathbb{P}\{Q\} = \mathbb{P}\{D_{w,k} > R_m\}$$

Step 3: Decomposing the Probability using Independence and Partitioning

Lemma 1.4 (Independence of Distance Components)**.** *For any fixed index i, the random variables $S_{2,i}$ and R_i are independent.*

Proof of Lemma. Let $v_i = p_i - q$. Since $p_i, q \sim \mathcal{N}(0, \Lambda)$ independently, $v_i \sim \mathcal{N}(0, 2\Lambda)$. Since Λ is diagonal, the components $v_{ij} = p_{ij} - q_j$ are independent across j, with $v_{ij} \sim \mathcal{N}(0, 2\lambda_j)$. $S_{2,i} = \sum_{j=1}^{d'} v_{ij}^2$ depends only on components $j = 1, \ldots, d'$. $R_i = \sum_{j=d'+1}^{d} v_{ij}^2$ depends only on components $j = d' + 1, \ldots, d$. Because the sets of components are disjoint and the components are independent, $S_{2,i}$ and R_i are independent random variables. □

While $S_{2,i}$ and R_i are unconditionally independent for a fixed i, the variables $D_{w,k}$ (which depends on the set $\{S_{2,1}, \ldots, S_{2,n}\}$ and their ranks) and R_m (which depends on $\{R_{r_1}, \ldots, R_{r_k}\}$ where the indices r_i depend on the S_2 ranks) are not necessarily independent. However, for the purpose of establishing a lower bound, we will work with the integral representation derived from conditioning or make approximations. Assuming the decomposition holds (possibly under approximation for large n or after conditioning and averaging):

$$\mathbb{P}(D_{w,k} > R_m) = \int_0^\infty \mathbb{P}(R_m \leq x) f_{D_{w,k}}(x) dx$$

where $f_{D_{w,k}}(x)$ is the probability density function (PDF) of $D_{w,k}$.

Introduce the partition $0 = d_0 \leq d_1 < d_2 < \cdots < d_{z-1} < d_z = +\infty$.

$$\begin{aligned}
\mathbb{P}(D_{w,k} > R_m) &= \sum_{i=1}^{z-1} \int_{d_i}^{d_{i+1}} \mathbb{P}(R_m \leq x) f_{D_{w,k}}(x) dx + \int_0^{d_1} \mathbb{P}(R_m \leq x) f_{D_{w,k}}(x) dx \\
&\geq \sum_{i=1}^{z-1} \int_{d_i}^{d_{i+1}} \mathbb{P}(R_m \leq x) f_{D_{w,k}}(x) dx \quad \text{(since the integral from 0 to } d_1 \text{ is non-negative)}
\end{aligned}$$

For $x \in (d_i, d_{i+1}]$, $\mathbb{P}(R_m \leq x) \geq \mathbb{P}(R_m \leq d_i)$ since the CDF is non-decreasing.

$$\begin{aligned}
\int_{d_i}^{d_{i+1}} \mathbb{P}(R_m \leq x) f_{D_{w,k}}(x) dx &\geq \mathbb{P}(R_m \leq d_i) \int_{d_i}^{d_{i+1}} f_{D_{w,k}}(x) dx \\
&= \mathbb{P}(R_m \leq d_i) \mathbb{P}(d_i < D_{w,k} \leq d_{i+1})
\end{aligned}$$

Therefore, we obtain the lower bound:

$$\mathbb{P}(D_{w,k} > R_m) \geq \sum_{i=1}^{z-1} \mathbb{P}(R_m \leq d_i)\mathbb{P}(d_i < D_{w,k} \leq d_{i+1}) \tag{3}$$

We now proceed to find lower bounds for the terms in this sum.

Step 4: Lower Bounding $\mathbb{P}\{R_m \leq d_i\}$

Recall $R_m = \max\{R_{r_1}, \ldots, R_{r_k}\}$. The indices $r_1, \ldots, r_k$ are random (depend on the S_2 ordering). However, by Lemma 1.4, $S_{2,j}$ and R_j are independent for each j. This implies that the set of values $\{R_{r_1}, \ldots, R_{r_k}\}$ has the same joint distribution as any fixed subset of k of the R_j's, say $\{R_1, \ldots, R_k\}$. This is because the selection criterion (smallest $S_{2,j}$) is independent of the R_j values. Therefore,

$$\mathbb{P}\{R_m \leq d_i\} = \mathbb{P}\{\max(R_{r_1}, \ldots, R_{r_k}) \leq d_i\} = \mathbb{P}\{\max(R_1, \ldots, R_k) \leq d_i\}$$

The variables $R_1, \ldots, R_k$ are identically distributed. They are conditionally independent given the query point q. Let $F_{R|q}(t) = \mathbb{P}(R_j \leq t|q)$ be the conditional CDF.

$$\begin{aligned}
\mathbb{P}\{\max(R_1, \ldots, R_k) \leq d_i\} &= \mathbb{P}\{R_1 \leq d_i, \ldots, R_k \leq d_i\} \\
&= \mathbb{E}_q[\mathbb{P}(R_1 \leq d_i, \ldots, R_k \leq d_i|q)] \\
&= \mathbb{E}_q[\prod_{j=1}^{k} \mathbb{P}(R_j \leq d_i|q)] \quad \text{(Conditional Independence)} \\
&= \mathbb{E}_q[(F_{R|q}(d_i))^k]
\end{aligned}$$

Let $X = F_{R|q}(d_i)$. Since $x \mapsto x^k$ is a convex function for $x \in [0,1]$ and $k \geq 1$, we apply Jensen's inequality:

$$\mathbb{E}_q[(F_{R|q}(d_i))^k] \geq (\mathbb{E}_q[F_{R|q}(d_i)])^k$$

By the law of total probability, $\mathbb{E}_q[F_{R|q}(d_i)] = \mathbb{P}(R_1 \leq d_i)$, which is the marginal CDF of R_1, denoted $F_R(d_i)$. Thus, $\mathbb{P}\{R_m \leq d_i\} \geq (F_R(d_i))^k$.

Now, we approximate the marginal distribution of $R_1 = \sum_{j=d'+1}^{d}(p_{1j} - q_j)^2$. Let $X_j = p_{1j} - q_j$. Since $p_{1j}, q_j \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \lambda_j)$, $X_j \sim \mathcal{N}(0, 2\lambda_j)$. Then $X_j^2/(2\lambda_j) \sim \chi_1^2$. So $R_1 = \sum_{j=d'+1}^{d}(2\lambda_j)Y_j$, where $Y_j \sim \chi_1^2$ are independent. This is a generalized chi-squared distribution. We use the Satterthwaite-Welch (SW) approximation defined in the Theorem statement (Item 1), matching the first two moments μ_R and σ_R^2. The approximate CDF is $F_R(d_i) \approx P(\nu_R/2, d_i/(2\theta_R))$.

Substituting this into the bound from Jensen's inequality:

$$\mathbb{P}\{R_m \leq d_i\} \gtrsim [F_R(d_i)]^k \tag{4}$$

where $F_R(d_i)$ is computed using the SW approximation. This provides the first factor for the sum in (3).

Step 5: Lower Bounding $\mathbb{P}\{d_i < D_{w,k} \leq d_{i+1}\}$

Let $Q_i = \mathbb{P}\{d_i < D_{w,k} \leq d_{i+1}\}$. We analyze this probability by conditioning on q.

$$Q_i = \mathbb{E}_q[\mathbb{P}(d_i < D_{w,k} \leq d_{i+1}|q)]$$

Distribution of $S_{2,i}|q$: Conditional on q, $S_{2,i}|q = \sum_{j=1}^{d'}(p_{ij} - q_j)^2$. Since $p_{ij} \sim \mathcal{N}(0, \lambda_j)$ independently, $p_{ij} - q_j \sim \mathcal{N}(-q_j, \lambda_j)$. Thus, $S_{2,i}|q$ is a sum of independent scaled non-central χ_1^2 variables: $S_{2,i}|q = \sum_{j=1}^{d'} \lambda_j \chi_1^2(q_j^2/\lambda_j)$. The exact conditional mean and variance are: $\mu_S(q) = \mathbb{E}[S_{2,i}|q] = \sum_{j=1}^{d'}(\lambda_j + q_j^2)$. $\sigma_S^2(q) = \text{Var}(S_{2,i}|q) = \sum_{j=1}^{d'}(2\lambda_j^2 + 4\lambda_j q_j^2) = Z(q)$.

Approximation for $S_{2,i}|q$ and $D_{w,k}|q$: We make the simplifying approximation that, for a fixed q and large n, the conditional distribution of $S_{2,i}|q$ can be treated as approximately normal for the purpose of analyzing the difference of order statistics:

$$S_{2,i}|q \approx \mathcal{N}(\mu_S(q), \sigma_S^2(q)) = \mathcal{N}(\mu_S(q), Z(q))$$

Under this assumption, and using standard large-n asymptotic results for order statistics from a normal sample (e.g., Bahadur-Kiefer representation), the difference $D_{w,k}|q = S_{2,(wk)}|q - S_{2,(k)}|q$ is approximately normally distributed:

$$D_{w,k}|q \overset{\text{approx}}{\sim} \mathcal{N}(\mu_D(q), \sigma_D^2(q))$$

The approximate parameters are derived using the quantiles of the approximated $S_{2,i}|q$ distribution:

$$\begin{aligned}
\mu_D(q) &\approx \mathbb{E}[S_{2,(wk)}|q] - \mathbb{E}[S_{2,(k)}|q] \\
&\approx (\mu_S(q) + \sigma_S(q)\Phi^{-1}(\beta_{wk})) - (\mu_S(q) + \sigma_S(q)\Phi^{-1}(\alpha_k)) \\
&= \sigma_S(q)(\Phi^{-1}(\beta_{wk}) - \Phi^{-1}(\alpha_k)) = \sqrt{Z(q)}C_{\alpha\beta}
\end{aligned}$$

$$\begin{aligned}
\sigma_D^2(q) &\approx \mathrm{Var}(S_{2,(wk)}|q - S_{2,(k)}|q) \\
&\approx \frac{\sigma_S^2(q)}{n}\left(\frac{\beta_{wk}(1-\beta_{wk})}{\phi(\Phi^{-1}(\beta_{wk}))^2} + \frac{\alpha_k(1-\alpha_k)}{\phi(\Phi^{-1}(\alpha_k))^2} - \frac{2\alpha_k(1-\beta_{wk})}{\phi(\Phi^{-1}(\alpha_k))\phi(\Phi^{-1}(\beta_{wk}))}\right) \\
&= \frac{Z(q)}{n}C_\sigma^2
\end{aligned}$$

Thus, the approximate standard deviation is $\sigma_D(q) \approx \sqrt{Z(q)}n^{-1/2}C_\sigma$.

Conditional Probability Approximation: Using the normality approximation for $D_{w,k}|q$:

$$\mathbb{P}(d_i < D_{w,k} \le d_{i+1}|q) \approx \Phi\left(\frac{d_{i+1} - \mu_D(q)}{\sigma_D(q)}\right) - \Phi\left(\frac{d_i - \mu_D(q)}{\sigma_D(q)}\right)$$

Defining Event $C_i(q)$ and Lower Bound η_i: Choose parameters $\delta_{i,1}, \delta_{i,2} \in \mathbb{R} \cup \{-\infty, +\infty\}$ with $\delta_{i,1} \le \delta_{i,2}$. Define the event $C_i(q)$ based on the standardized thresholds relative to the approximate distribution of $D_{w,k}|q$:

$$C_i(q): \quad \frac{d_i - \mu_D(q)}{\sigma_D(q)} \le \delta_{i,1} \quad \text{and} \quad \frac{d_{i+1} - \mu_D(q)}{\sigma_D(q)} \ge \delta_{i,2}$$

If event $C_i(q)$ occurs, the conditional probability is lower bounded (under the normality approximation):

$$\mathbb{P}(d_i < D_{w,k} \le d_{i+1}|q) \approx \Phi\left(\frac{d_{i+1} - \mu_D(q)}{\sigma_D(q)}\right) - \Phi\left(\frac{d_i - \mu_D(q)}{\sigma_D(q)}\right) \ge \Phi(\delta_{i,2}) - \Phi(\delta_{i,1}) =: \eta_i$$

Note that $\eta_i \ge 0$.

Lower Bounding Q_i: Using the law of total expectation and the indicator function $\mathbb{I}(C_i(q))$:

$$\begin{aligned}
Q_i &= \mathbb{E}_q[\mathbb{P}(d_i < D_{w,k} \le d_{i+1}|q)] \\
&\ge \mathbb{E}_q[\mathbb{P}(d_i < D_{w,k} \le d_{i+1}|q)\mathbb{I}(C_i(q))] \quad \text{(term is non-negative)} \\
&\gtrsim \mathbb{E}_q[\eta_i \mathbb{I}(C_i(q))] \quad \text{(using the lower bound } \eta_i \text{ when } C_i(q) \text{ occurs)} \\
&= \eta_i \mathbb{P}_q(C_i(q))
\end{aligned}$$

The approximation $\gtrsim$ depends on the accuracy of the normality approximation for $D_{w,k}|q$.

Analyzing the Event $C_i(q)$: Substitute the approximations $\mu_D(q) \approx \sqrt{Z(q)}C_{\alpha\beta}$ and $\sigma_D(q) \approx \sqrt{Z(q)}n^{-1/2}C_\sigma$. The conditions defining $C_i(q)$ become:

1. $d_i - \sqrt{Z(q)}C_{\alpha\beta} \le \delta_{i,1}\sqrt{Z(q)}n^{-1/2}C_\sigma$
2. $d_{i+1} - \sqrt{Z(q)}C_{\alpha\beta} \ge \delta_{i,2}\sqrt{Z(q)}n^{-1/2}C_\sigma$

Rearranging to isolate $\sqrt{Z(q)}$ (assuming $Z(q) > 0$):

1. $d_i \le \sqrt{Z(q)}(C_{\alpha\beta} + \delta_{i,1}n^{-1/2}C_\sigma) = \sqrt{Z(q)}C_{tot}(\delta_{i,1})$
2. $d_{i+1} \ge \sqrt{Z(q)}(C_{\alpha\beta} + \delta_{i,2}n^{-1/2}C_\sigma) = \sqrt{Z(q)}C_{tot}(\delta_{i,2})$

Assume $d_i > 0$. For the inequalities to constrain $Z(q)$ non-trivially, we require $C_{tot}(\delta_{i,1}) > 0$ and $C_{tot}(\delta_{i,2}) > 0$. The conditions become:

1. $\sqrt{Z(q)} \ge d_i/C_{tot}(\delta_{i,1})$
2. $\sqrt{Z(q)} \le d_{i+1}/C_{tot}(\delta_{i,2})$ (if $d_{i+1} < \infty$)

Combining these, the event $C_i(q)$ is approximately equivalent to the event:

$$\frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2} \le Z(q) \le \frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2}$$

If $d_{i+1} = \infty$ (which occurs for $i = z-1$), the second inequality $\sqrt{Z(q)} \le \infty/C_{tot}(\delta_{i,2})$ is always true (given $C_{tot}(\delta_{i,2}) > 0$). In this case, the event is just $Z(q) \ge d_i^2/[C_{tot}(\delta_{i,1})]^2$. Note that choosing $\delta_{i,2} = +\infty$ makes the second condition $d_{i+1} - \mu_D(q) \ge \infty \times \sigma_D(q)$ which requires $d_{i+1} = \infty$ if $\sigma_D(q)$ is finite and positive. So setting $d_{i+1} = \infty$ naturally aligns with setting $\delta_{i,2} = +\infty$.

Calculating $\mathbb{P}_q(C_i(q))$: We approximate the probability of this event using the distribution of $Z(q)$. The distribution of $Z(q) = \sum_{j=1}^{d'}(2\lambda_j^2 + 4\lambda_j q_j^2)$ depends on $q \sim \mathcal{N}(0, \Lambda)$. $Z(q)$ involves a sum of scaled χ_1^2 variables q_j^2/λ_j. We use the SW approximation for $Z'(q) = Z(q) - \sum 2\lambda_j^2$, as defined in the Theorem statement (Item 2). Let $\bar{F}_Z(y) = \mathbb{P}_q(Z(q) \ge y)$ be the approximate survival function derived from the SW approximation for $Z'(q)$. Then,

$$\begin{aligned}
\mathbb{P}_q(C_i(q)) &\approx \mathbb{P}_q\left(\frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2} \le Z(q) \le \frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2}\right) \\
&= \mathbb{P}_q\left(Z(q) \ge \frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2}\right) - \mathbb{P}_q\left(Z(q) > \frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2}\right) \\
&\approx \bar{F}_Z\left(\frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2}\right) - \bar{F}_Z\left(\frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2}\right)
\end{aligned}$$

The second term is 0 if $d_{i+1} = \infty$. Let this approximation be denoted $L_{C,i}$. Note that $L_{C,i} \ge 0$ requires $\frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2} \le \frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2}$ assuming $\bar{F}_Z$ is non-increasing.

Lower Bound for Q_i: Combining the results:

$$Q_i = \mathbb{P}\{d_i < D_{w,k} \le d_{i+1}\} \gtrsim \eta_i \mathbb{P}_q(C_i(q)) \approx \eta_i L_{C,i}$$

$$Q_i \gtrsim [\Phi(\delta_{i,2}) - \Phi(\delta_{i,1})]\left[\bar{F}_Z\left(\frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2}\right) - \bar{F}_Z\left(\frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2}\right)\right] \tag{5}$$

Let $L_{Q,i}$ denote this lower bound for Q_i. Note $L_{Q,i} \ge 0$.

Step 6: Combining Bounds

Substitute the lower bounds for $\mathbb{P}(R_m \le d_i)$ (from Step 4, denoted $L_{P,i} = [F_R(d_i)]^k$) and $\mathbb{P}(d_i < D_{w,k} \le d_{i+1})$ (from Step 5, denoted $L_{Q,i}$) into the inequality (3) from Step 3:

$$\mathbb{P}\{T_1 \subseteq T_2\} \ge \mathbb{P}(D_{w,k} > R_m) \ge \sum_{i=1}^{z-1} \mathbb{P}(R_m \le d_i)\mathbb{P}(d_i < D_{w,k} \le d_{i+1})$$

Using the derived lower bounds $L_{P,i}$ and $L_{Q,i}$, which are non-negative:

$$\mathbb{P}\{T_1 \subseteq T_2\} \gtrsim \sum_{i=1}^{z-1} L_{P,i} L_{Q,i}$$

$$= \sum_{i=1}^{z-1} [F_R(d_i)]^k \times [\Phi(\delta_{i,2}) - \Phi(\delta_{i,1})] \left[\bar{F}_Z \left(\frac{d_i^2}{[C_{tot}(\delta_{i,1})]^2} \right) - \bar{F}_Z \left(\frac{d_{i+1}^2}{[C_{tot}(\delta_{i,2})]^2} \right) \right]$$

This is the final result stated in equation (1) of the theorem, using $\eta_i = \Phi(\delta_{i,2}) - \Phi(\delta_{i,1})$.

The derivation of the special case for $z = 2$ follows directly by setting $i = 1$, $d_1 = \mu_R + y\sigma_R$, $d_2 = \infty$, $\delta_{1,1} = \delta$, and $\delta_{1,2} = +\infty$ as justified in the theorem statement and Step 5.

□

Remark 1.5 (Summary of Approach and Approximations)**.** The proof establishes a lower bound for $\mathbb{P}\{T_1 \subseteq T_2\}$ based on the following key steps and approximations:

1. Identifying the sufficient condition $D_{w,k} > R_m$.
2. Decomposing $\mathbb{P}(D_{w,k} > R_m)$ over a partition $d_1, \ldots, d_{z-1}$ using the law of total probability, leading to the sum $\sum \mathbb{P}(R_m \leq d_i)\mathbb{P}(d_i < D_{w,k} \leq d_{i+1})$.
3. Bounding $\mathbb{P}(R_m \leq d_i)$ using Jensen's inequality on the conditional probabilities $\mathbb{E}_q[(F_{R|q}(d_i))^k]$ and approximating the marginal CDF $F_R(d_i)$ using the Satterthwaite-Welch (SW) method for the generalized χ^2 distribution of R_1.
4. Bounding the interval probability $\mathbb{P}(d_i < D_{w,k} \leq d_{i+1})$ via conditional expectation:
   - Approximating the conditional distribution $D_{w,k}|q$ as Normal $\mathcal{N}(\mu_D(q), \sigma_D^2(q))$ for large n, where $\mu_D(q)$ and $\sigma_D^2(q)$ depend on the conditional variance $Z(q)$.
   - Defining an event $C_i(q)$ such that if it occurs, the conditional probability $\mathbb{P}(d_i < D_{w,k} \leq d_{i+1}|q)$ is lower bounded by a constant $\eta_i = \Phi(\delta_{i,2}) - \Phi(\delta_{i,1})$.
   - Relating the event $C_i(q)$ to a range constraint on the conditional variance $Z(q)$.
   - Approximating $\mathbb{P}_q(C_i(q))$ using the SW approximation for the distribution of $Z(q)$.
   - Combining these to yield $Q_i \gtrsim \eta_i \mathbb{P}_q(C_i(q))$.
5. Combining the bounds for the terms in the sum.

The final bound's quality depends on the accuracy of the large-n order statistic asymptotics, the conditional normality approximation for $D_{w,k}|q$, and the SW approximations for the distributions of R_1 and $Z(q)$. The parameters z, y_i, $\delta_{i,1}$, $\delta_{i,2}$ provide flexibility in optimizing the bound. The special case $z = 2$ offers a simpler application with parameters y and δ. This theorem provides theoretical backing for the intuition that reducing dimensionality with PCA and over-selecting candidates (wk) can preserve a high probability of capturing the true nearest neighbors, especially when the variance captured by the first d' components is significant relative to the residual variance.